Thibault Pautrel



Postdoctoral researcher in applied mathematics and computer science

Riemannian optimization and federated learning

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Born 17 August 1994 in Le Mans, France — French citizen

PROFESSIONAL EXPERIENCE

2025–present
Postdoctoral researcher
CentraleSupélec, Laboratory of Signals and Systems (L2S) — Gif-sur-Yvette

Project: Federated learning on covariance matrices for pre-trained models on health data.
Supervision: Florent Bouchard.

2025
6 months
Research internship (MSc)
LISTIC, Université Savoie Mont Blanc — Annecy

Project: Riemannian Federated Learning on remote sensing data.
Supervision: Guillaume Ginolhac, Ammar Mian, Florent Bouchard.

2021–2024
Full time
Mathematics teacher (*agrégé*)
French national education system

Teaching in secondary schools. Weekly oral examinations in undergraduate preparatory classes (BCPST, PCSI, MP).

2018–2021
PhD in Mathematics
Université de Rennes 1, IRMAR

Title: Some contributions to the study of the universality of the zeros of random trigonometric functions (defended 10 January 2022).
Advisors: Jürgen Angst and Guillaume Poly.
Funding: Doctoral contract with a teaching assignment (64 h/year).

2018
5 months
Research internship (MSc)
LAMA, Université Savoie Mont Blanc — Chambéry

Project: Mean-reflected backward stochastic differential equations.
Supervision: Philippe Briand.

PUBLICATIONS

Computer science (artificial intelligence & geometric learning)

Keywords — Federated learning, Riemannian manifolds, SPD matrices, geometric deep learning, optimization on manifolds, signal processing.

[C4] Submitted — **T. Pautrel**, F. Bouchard, G. Ginolhac, A. Mian, A. Bellet. *Projection-based federated learning on compact submanifolds with geometry-independent differential privacy*. Submitted to TMLR, 2026.

[C3] Submitted — **T. Pautrel**, F. Bouchard, A. Mian, G. Ginolhac. *Personalised federated learning for Riemannian and Euclidean EEG decoding*. Submitted to ICASSP 2027. arXiv:2609.29037.

[C2] Accepted — **T. Pautrel**, F. Portier. *Riemannian stochastic optimization for sufficient dimension reduction*. ICML 2026

[C1] Accepted — **T. Pautrel**, F. Bouchard, A. Mian, G. Ginolhac. *FedSPDnet: geometry-aware federated deep learning with SPDnet*. EuSIPCO 2026.

Mathematics (stochastic processes & asymptotic analysis)

Keywords — Nodal sets, random trigonometric polynomials, stationary Gaussian processes, strong dependence, universality, Kac–Rice formulas.

[R2] To appear — J. Angst, **T. Pautrel**, G. Poly. *Global universality of the expected number of zeros of non-analytic random signals*. INdAM–Springer Proceedings, 2025.

[R1] Published — J. Angst, **T. Pautrel**, G. Poly. *Real zeros of random trigonometric polynomials with dependent coefficients*. Transactions of the American Mathematical Society **375** (2022), 7209–7260.

[R0] Published — **T. Pautrel**. *New asymptotics for the mean number of zeros of random trigonometric polynomials with strongly dependent Gaussian coefficients*. Electronic Communications in Probability **25** (2020), no. 36.

TEACHING

University teaching

2026–2027 · 6 h TD-eq.

Labs: Machine Learning for Data Science
ENSAI — MSc Smart Data
17 students, 6h.

2025–2027 · 33 h TD-eq.

Labs: Introduction to C++ programming
IUT Orsay — 1st year (R101-B)
15 students, 5.5 h per week.

2025–2026 · 12 h TD-eq.

Tutorials and labs: Machine Learning
CentraleSupélec — 2nd year
102 students, taught in English, 8 sessions of 1.5 h.

2025–2026

Lectures and tutorials: Statistics and Learning
CentraleSupélec — 1st year
≈30 students, taught in French and English (S2).
Lectures: $2\times1.5\,\text{h} = 4.5\,\text{h}$ TD-eq. + 15 h of tutorials.

2025–2026 · 1.5 h TD-eq.

Tutorials: Digital image processing
CentraleSupélec — 2nd year
Taught in French.

2018–2021 · 48 h TD-eq.

Tutorials: Analysis and probability
Université de Rennes 1 — 2nd year, Mathematics
≈30 students, 4 h per week.

2018–2021 · 16 h TD-eq.

Tutorials: Probability
Université de Rennes 1 — 1st year, Biology
≈30 students, 2 h per week over 8 weeks.

Supervision and examination boards

2026

Master's thesis examination board
ENSAI — MSc Smart Data

2023–2025

Examiner, CentraleSupélec entrance examination
Marker for the written MP Mathematics 1 papers (2023, 2024, 2025).
Examiner for the Mathematics–Computer Science oral examinations (2023, 2024).

2019–2020

Supervision of undergraduate research projects
ENS Rennes — 1st year
Topic: Stein's method for normal approximation.

2018–2021

Mock oral examination board
Université de Rennes 1
Preparation for the CAPES and the *agrégation* in mathematics.

Oral examinations in undergraduate preparatory classes

2021–2024

Weekly oral examinations in mathematics
BCPST, PCSI, MP — various institutions

2016–2018 **Weekly oral examinations in mathematics**
MP, MP* — Lycée Chateaubriand, Rennes

Secondary school teaching

2021–2024
Full time
Mathematics teacher (*agrégé*)
Secondary schools

EDUCATION

2024–2025 **MSc Smart Data**
ENSAI, Rennes — with highest honours
Specialisation: machine learning, deep learning, dimension reduction, functional data analysis.

2017–2018 **MSc in Probability and Statistics ("Aléatoire")**
Université de Rennes 1 — with honours
Specialisation: stochastic processes and stochastic calculus, statistics of processes, dynamical systems, parametric and non-parametric statistics, Malliavin calculus and Stein's method

2016–2017 **Preparation for the *agrégation* in mathematics**
Université de Rennes 1 — with honours

2015–2016 **MSc (first year) in pure mathematics**
Université de Rennes 1 — with honours

2014–2015 **BSc in mathematics**
Université de Rennes 1 — with honours

2012–2014 **Undergraduate preparatory classes, MPSI – MP***
Lycée Chateaubriand, Rennes

COMPETITIVE EXAMINATIONS AND DISTINCTIONS

2026 **National qualification for Associate Professor positions**
CNU section 26 — applied mathematics and applications of mathematics

2017 ***Agrégation externe* in mathematics**
Probability–statistics option — rank: 118

SERVICE

2026 **Reviewer**
ICML 2026 — International Conference on Machine Learning

2019–2021 **Co-organiser of the doctoral seminar**
"Gaussbusters" seminar — IRMAR, Rennes
Weekly seminar of the PhD students in probability and statistics.

TECHNICAL SKILLS

Languages	Python (NumPy, PyTorch, scikit-learn), R, SQL, C++, OCaml, Scilab
Computing	Distributed computing (Spark), parallel computing, HTCondor scheduler
Tools	Git, LaTeX, Unix/Bash

LANGUAGES

French	Native
English	Fluent (C2) *Certificate of Proficiency in English (Cambridge, 2013)*
Spanish	Intermediate (B2) *Read, written, spoken*
Breton	Beginner